



Skill-Based Assessment -

Module 331: Excel Foundations - TWO

Objective:

In this skill-based assessment (SBA), you are expected to analyze and visualize the given dataset using Microsoft Excel.

Time/Duration:

- You have **24 hours** to complete this SBA, and your instructor will designate at least 5.5 hours for you to work on it in class.
- Please make sure that you are not interrupted during the test.

Passing Criteria

- **Total Points: 100.**
Passing Score: 70% (70 points).

Instructions

Sample Dataset:

→ [Click here to download the Excel file for the SBA](#)



Section One: Excel Formula and Function

(60 points)

Below are the tasks:

- 1.1. Convert your sheet to a table, and the table name should be **EmployeeTable**.
 - a. *Note: Take a screenshot of your sheet before and after the conversion, and submit the screenshots of the worksheet.*
- 1.2. Create a new column containing **email IDs** for all employees, if one does not already exist, using the format **firstname.lastname@company.com**
- 1.3. Extract the first **4 letters** of each employee's first name and create a new column named **Emp_Initial_Code**.
- 1.4. Create a new column named **Region Code**, if it does not already exist. Then, use the **SUBSTITUTE()** function to replace **"North"** with **"N"**, **"East"** with **"E"**, **"South"** with **"S"**, and **"West"** with **"W"** in the Region column.
- 1.5. Create a new column named **EM_Code** if it does not already exist. The **EM_Code** represents the **Employee Identifier**, which you can create by concatenating the following columns:
 - Region Code.
 - Employee ID.
 - First letter of **"First name"**
 - First letter of **"Last name"**:
 - Use **CONCATENATE()** or the modern **TEXTJOIN()**.



- 1.6. Create a new column named **"Joining Date"** if it does not already exist. Format the **Joining date** in U.S. format (**MM-DD-YYYY**) (Hint: TEXT() function).
- 1.7. Create a new column named **"Promotion Eligibility"**, if it does not already exist, and determine whether each employee is **`eligible`** or **`not eligible`** for promotion based on the following criteria:
 - a. **Criteria:** Performance Rating ≥ 4.0 and Bonus% $\geq 10\%$.
- 1.8. Replace the **area code** of the employee from **929** to **323**.
- 1.9. Use **INDEX()** and **MATCH()** to find the **`Region`** of the employee with the highest performance rating.
- 1.10. Calculate the number of **years** each employee has been with the company (based on their **joining date**).
- 1.11. Calculate the **total salary for all employees** by summing the values in the **Salary** column (excluding bonuses).
- 1.12. Find the **average bonus percentage** across all employees.
- 1.13. Find the **maximum salary** in the dataset.
- 1.14. Determine the **standard deviation** of the performance ratings.
- 1.15. Calculate the **standard deviation** of the **`Monthly Sales`** for all employees.
- 1.16. Determine the **variance** of the **`Monthly Sales.`**
- 1.17. Find the **median** of the **`Monthly Sales.`**
- 1.18. Find the **mode** of the **`Monthly Sales.`**
- 1.19. Calculate the **Monthly Loan Payment (PMT)** based on:
 - **Loan Amount:** Column Value.
 - **Interest Rate:** Annual rate divided by 12.
 - **Loan Term:** Column value in months.



Section Two: PivotTable and PivotChart

(40 points)

2.1. Salary and Performance Analysis by Department

2.1.1. Create a PivotTable to summarize employee data by **Department**:

- Calculate the **Total Salary**.
- Calculate the **Average Performance Rating**.
- Count the **number of employees**.

2.1.2. Create a PivotChart to visualize the summarized data from the PivotTable:

- Use a **Bar Chart** to display the **Total Salary** by department.

2.1.3. Add a slicer for **Region** to enable interactive filtering of the PivotTable.

2.2. Loan Analysis by Department

2.2.1. Create a PivotTable for loan analysis by department:

- Identify which department has the highest total loan amount.
- Determine which department has the lowest average interest rate.

(Hint: Add Sum of Loan Amount to Values and Average Interest Rate to Values.)

2.2.2. Create a combo chart to visualize the above requirements.

2.3. Employee Tenure

2.3.1. Create a PivotTable to analyze **'Employee Tenure:'**

- Identify which region has the **most experienced employees**.
- Determine which region has the **least number of employees**.

Hint: Use **Years with Company** in Values (summarized as **Average**).

2.3.2. Create a **'Clustered Column Chart'** to visualize the average employee tenure and employee count by region.

2.4. Tenure vs Performance

2.4.1. Create a PivotTable to calculate the following:

- **Average Years with Company** for each department.
- **Average Performance Rating** for each department.

2.4.2. Create a **Line Chart** to visualize the relationship:

- Plot **Average Years with Company** and **Average Performance Rating** for each department.

SBA Submission Guidelines

- Save your work frequently to prevent data loss.
- Ensure all formulas remain intact in your submission (no hardcoding of values).
- Save your work in the required file format:
 - Excel files should be in **`.xlsx`** format.



- Screenshots or images should be in **.png** or **.jpeg** format.
- Name your file in the following format:
YourName_SBATitle.xlsx
Example: **JohnDoe_SBAM331.xlsx**
- Include formulas or functions or any technical details in a **Word document** with instructions.
- Submit the **final workbook** and all related files to the Canvas assignment submission.
- Late submissions may not be accepted.

Formulas and Functions Submission Guidelines

- Ensure that the formulas are applied in the correct columns as specified in the questions.
- Double-check that all functions are working as intended and returning accurate results.
- Avoid hardcoding values; use cell references in your formulas.
- Ensure all formulas are intact and not replaced with hardcoded values.
- Include a separate sheet, if necessary, to show intermediate calculations or explanations.
- You can add comments or a helper sheet to document the purpose of each formula you have used.



Submission Requirements

- **PivotTables:** Ensure all requested PivotTables are created and properly formatted.
 - Include clear labels and meaningful values in the PivotTable.
- **PivotCharts:** All requested charts must be included and labeled correctly.
 - Add a chart title, axis labels, and any necessary legends.
- **Screenshots:**
 - Take a **before and after** screenshot of your worksheet/table, where applicable.
 - Ensure the screenshots are clear and show the necessary details.

Comments and Notes

- Use comments or text boxes in Excel to provide any necessary explanations for your analysis.
- Highlight specific areas in your worksheet if required.